

Supplement: Egress thresholds and wildfire fatalities

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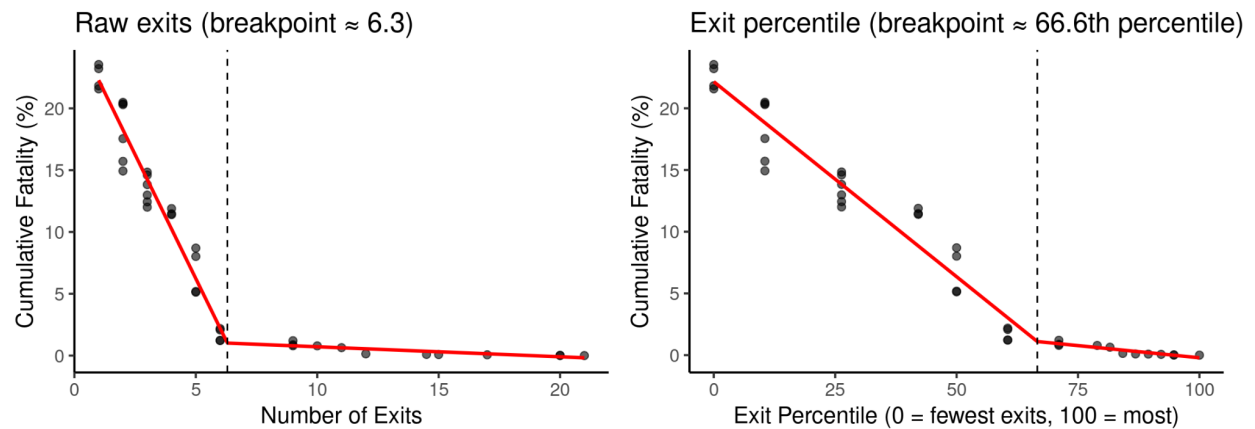


Figure S1. Segmented regression of cumulative wildfire fatalities versus community egress. **(a)** Cumulative per-capita fatalities plotted against the raw number of road exits per community, with a segmented (“broken-stick”) regression (red line) identifying a breakpoint at ~6.3 exits (dashed line). **(b)** The same cumulative fatalities plotted against exit percentiles (0–100), controlling for the uneven distribution of exit counts, with a corresponding segmented regression breakpoint at the 66.6th percentile (6 exits, dashed line). Percentile-based analysis provides a relative, data-normalized perspective, while the raw exit counts maintain physically interpretable values. Communities near the breakpoint experience the greatest marginal change in cumulative fatalities, highlighting the threshold at which additional egress has diminishing benefit.

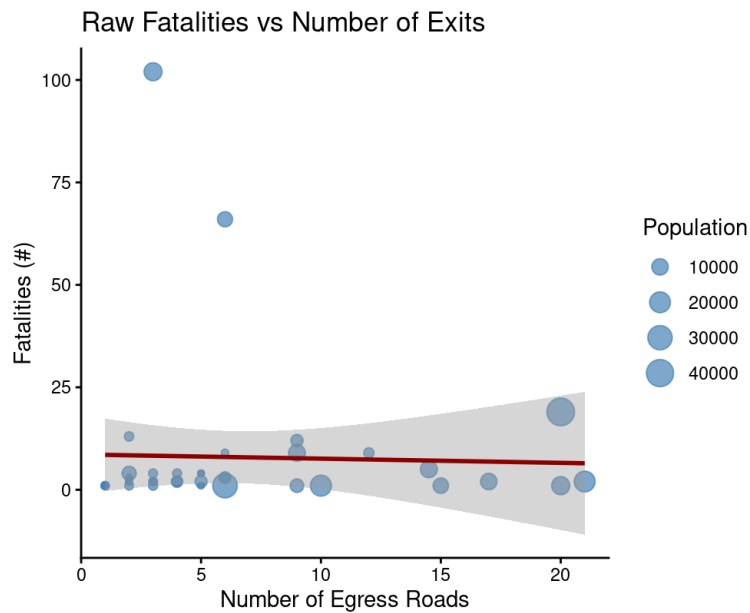


Figure S2. Raw wildfire fatalities versus the number of egress roads per community. Each point represents a community, with point size proportional to population. A linear regression (red line) shows the overall trend between number of exits and total fatalities. Larger communities tend to have both more exits and more fatalities, highlighting that raw fatalities are strongly influenced by population. Linear regression indicates no significant association (slope = -0.089 , $p = 0.87$, $R^2 = 0.0007$), motivating the use of per-capita metrics to compare communities of different sizes.

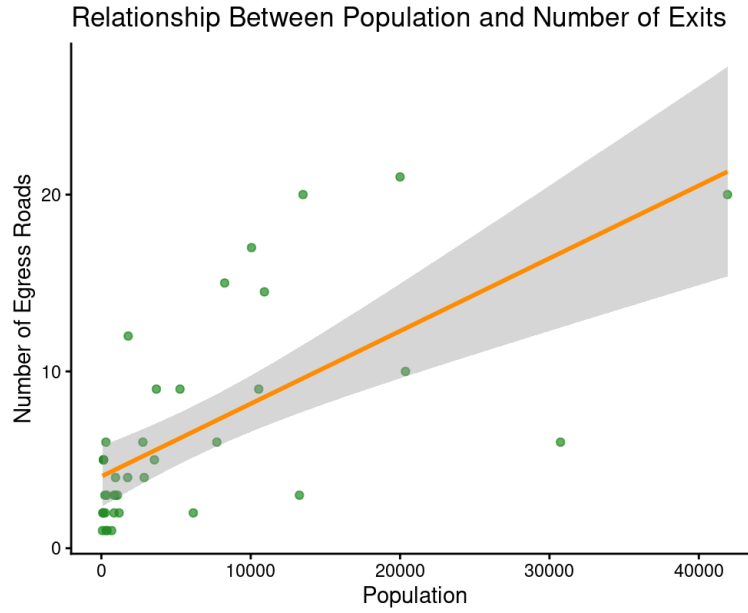


Figure S3. Relationship between community population and number of egress roads. Each point represents a community, and the line shows a linear regression between population and exits. As expected, larger communities tend to have more exits, reflecting planning and infrastructure scaling with population. Linear regression confirms a strong positive association (slope = 1038.6, $p = 6.03 \times 10^{-6}$, $R^2 = 0.429$), highlighting the link between community size and road infrastructure.

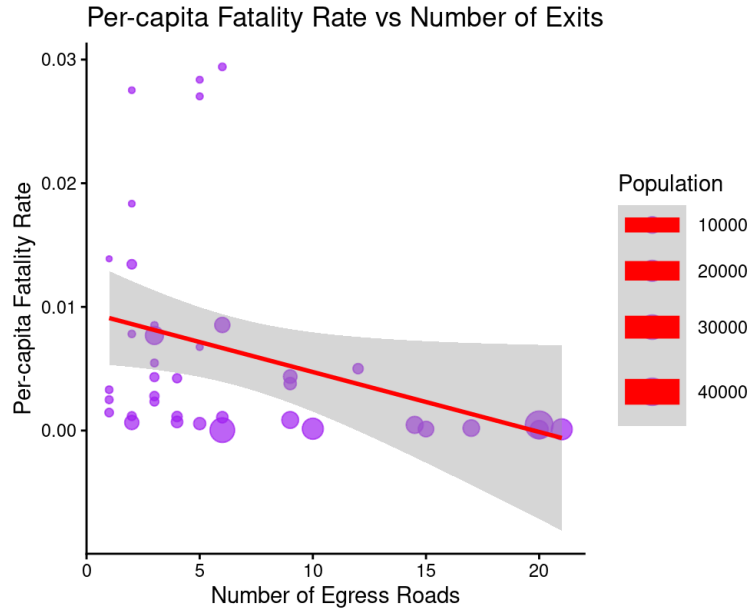


Figure S4. Per-capita wildfire fatality rate versus the number of egress roads per community. Fatalities are normalized by population to control for differences in community size. Each point represents a community, with point size proportional to population. A linear regression (red line) indicates that communities with fewer exits tend to have higher per-capita fatality rates, supporting the threshold pattern identified in the main text. Linear regression confirms this negative association (slope = -0.000485 , $p = 0.048$, $R^2 = 0.10$), demonstrating that reduced egress infrastructure is associated with increased relative risk of fatalities across communities of varying size.

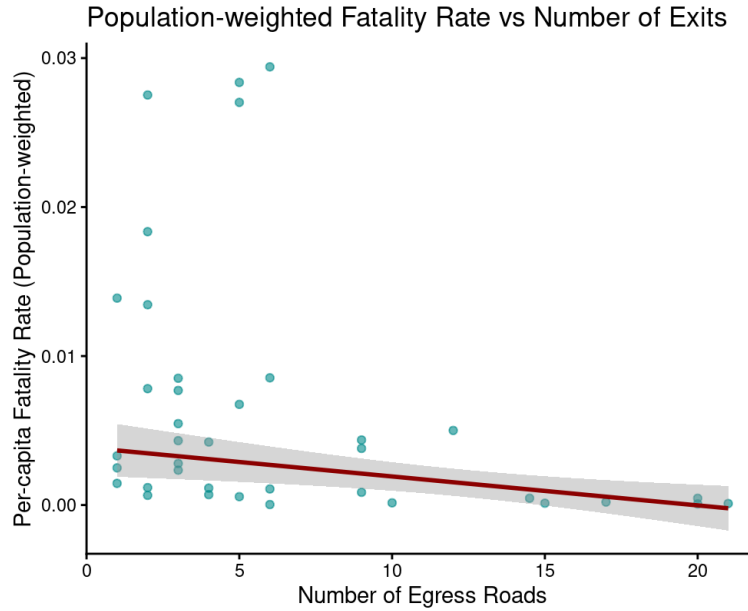


Figure S5. Population-weighted per-capita fatality rate versus the number of egress roads.

Each point represents a community, and the linear regression (red line) is weighted by population to emphasize larger communities, which contribute more to overall wildfire fatalities. Weighting confirms that the threshold relationship between exits and per-capita fatalities persists, showing that the main pattern is robust to population differences and is not driven solely by small communities with extreme rates. Weighted linear regression results indicate a negative slope (slope = -0.000194 , $p = 0.006$, $R^2 = 0.19$), highlighting the persistent association between reduced egress infrastructure and elevated per-capita risk.

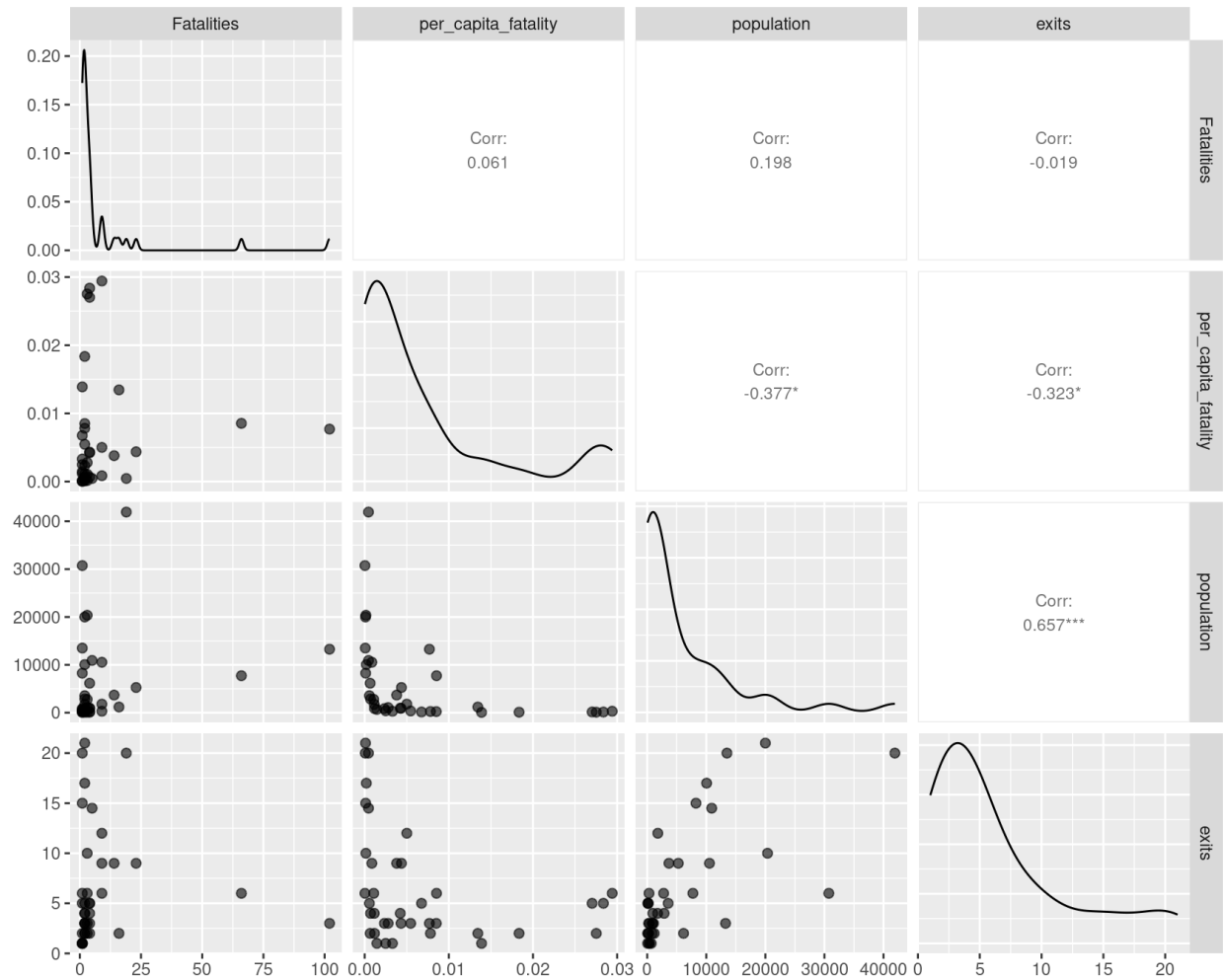


Figure S6. Pairwise relationships among key variables: total fatalities, per-capita fatalities, population, and number of exits. The diagonal shows the distribution of each variable, the lower triangle shows scatterplots of variable pairs, and the upper triangle displays Spearman correlation coefficients. As expected, population is positively correlated with total fatalities and number of exits. Per-capita fatalities show a negative correlation with exits, consistent with the threshold effect observed in the main analysis. This figure illustrates both the structural correlations in the dataset and the robustness of the per-capita metric for comparing communities of different sizes.

Table S1. **Top 20 highest-risk communities.** See supplement for all Census Places under 50,000.

Risk	Place	State	Hazard	Vulnerability	Exposure
1	Eagle Mountain	Utah	0.93	0.92	1.00
2	Sierra Vista	Arizona	1.00	0.89	1.00
3	Titusville	Florida	0.82	0.80	1.00
4	Tooele	Utah	0.84	0.93	0.93
5	Altadena	California	0.97	0.77	1.00
6	Wildomar	California	0.94	0.78	0.96
7	Rancho Santa Margarita	California	0.94	0.75	1.00
8	Poway	California	0.92	0.69	1.00
9	Parkland	Florida	0.64	0.76	0.89
10	Claremont	California	1.00	0.71	0.96
11	Temescal Valley	California	0.88	0.92	0.73
12	La Verne	California	1.00	0.77	0.83
13	Rockledge	Florida	0.80	0.86	0.72
14	Wenatchee	Washington	0.85	0.68	0.93
15	El Dorado Hills	California	0.69	0.83	1.00
16	Port St. John	Florida	1.00	0.91	0.65
17	Rio Rico	Arizona	0.97	0.93	0.55
18	Ramona	California	1.00	0.88	0.58
19	Elko	Nevada	1.00	0.93	0.54
20	Fallbrook	California	0.62	0.88	0.87

Supplementary Information: Data and code legends

Repository: <https://github.com/WRI-Science/roads-pnas>

R version: R 4.4.0 — tested with the package versions noted under each script.

1. fire fatalities with notes.xlsx (Microsoft Excel workbook)

Description

Curated, fire-level and place-level records used to document wildfire-related fatalities, link counts to U.S. Census places, and retain traceability to public sources. This workbook is the **authoritative tabular source** for fatality documentation; the CSV file below is a simplified export for scripted analysis.

Structure

The workbook contains **two worksheets**:

Worksheet	Content
redbooks + top 20 fires	Main table of fires/places with fatality counts, community attributes, and source documentation.
Tubbs	Additional documentation for the Tubbs fire (detail separated from the main sheet).

Variables (main sheet; column order as in file)

Column (file header)	Description
Source	Provenance label for the record (e.g. official fire agency compilation, curated major events).
Year	Calendar year of the fire event.
Fire	Fire or incident name.
Fatalities	Reported fatality count for the incident as encoded in this row (may differ from place-attributed counts).
notes	Free-text documentation (e.g. names, ages, circumstances) used for verification and transparency. May contain identifiable information.
fatality information	URLs or citations to news, agency, or other public reporting supporting the entries.
Fatalities that could not be assigned to a CP	Count or note for deaths that could not be assigned to a specific Census place (CP).
Identified fatalities	Count of fatalities attributed to the listed census place for analysis.
census place	U.S. Census place name associated with the attributed fatalities.
population	Place population used for rate calculations in scripts (see manuscript methods for source).

exits	Count of modeled or summarized egress road connections for that place (see main paper or data pipeline for definition).
evac_vul	Evacuation-vulnerability-related quantity as recorded in the workbook (not present in the CSV export used by the R scripts).

Notes

Links point to third-party public sources; URL availability may change after publication. We have copies of all urls, available upon request.

2. fire-fatality-for-R.csv (comma-separated values)

Description

Comma-delimited export of the fatality–place table **without** the `evac_vul` column, formatted for use in R with `read.csv()`.

Record unit

One row per **fire × census place** aggregation used in the fatality–egress analyses.

Variables

Same columns as in the Excel main sheet **except** `evac_vul` is omitted:

Source, Year, Fire, Fatalities, notes, fatality information, Fatalities that could not be assigned to a CP, Identified fatalities, census place, population, exits

R naming convention

`read.csv()` converts spaces and some characters to dots. Examples:

- Identified fatalities → Identified.fatalities
- census place → census.place
- fatality information → fatality.information
- Fatalities that could not be assigned to a CP → Fatalities.that.could.not.be.assigned.to.a.CP

Relationship to other files

Read by `fatalities.R` and `fire_fatalities_supplement.R`. The Excel workbook is the richer, human-audited source; this CSV is the analysis input.

3. fatalities.R (R script)

Description

Script for **segmented (piecewise) regression** of **cumulative place-level fatality share** as a function of **number of exits**, after filtering to complete cases. Produces a diagnostic plot (points, fitted segmented line, estimated breakpoint).

Primary input

- fire-fatality-for-R.csv (the read.csv() path in the script is machine-specific and must be set locally).

Processing (summary)

- Filters to rows with non-missing Identified.fatalities, exits, and population.
- Computes per_fatal = identified fatalities per 100 population; sorts by exits; forms **reversed cumulative** cum_per_fatal for the y-axis.
- Fits `lm(cum_per_fatal ~ exits)` then `segmented()` for one breakpoint; plots data and fitted curve; fits `lm(exits ~ population)` as an auxiliary check.

Outputs

- Figures sent to the **active R graphics device** (the distributed script does not call ggsave). Users may save plots from the IDE or add ggsave locally.

Software

R packages dplyr, ggplot2, segmented

R version *Insert tested version at submission.*

4. fire_fatalities_supplement.R (R script)

Description

Supplementary / exploratory fatality analyses extending the core segmented-regression workflow: fits on **raw exits** and **percentile-ranked exits** (side-by-side panels), plus additional bivariate and multivariate summaries (linear models, population weighting, correlation panel).

Primary input

- fire-fatality-for-R.csv (path in script is machine-specific).

Processing (summary)

- Same cleaning as fatalities.R, plus exits_pct (percentile rank of exits).
- Two segmented regressions (raw vs percentile x).

- Scatter plots with `geom_smooth(method = "lm")` for: identified fatalities vs exits; population vs exits; per-capita fatality rate vs exits (including population-weighted regression).
- Pairwise plot via `GGally::ggpairs` on selected numeric columns.

Outputs

- Multiple **ggplot2** objects (printed to the graphics device). The repository copy reviewed does not include bundled ggsave paths.

Software

R packages `dplyr`, `ggplot2`, `segmented`, `GGally`, `patchwork`

R version *Insert tested version at submission.*

5. pnas-mapping.R (R script)

Description

Main manuscript mapping and composite-risk workflow at **U.S. community (place) points**: reads the frozen merged roads/evacuation/wildfire dataset, merges state boundaries and state population from the Census API, derives exits/lanes and normalized hazard/exposure/evacuation vulnerability, assigns bins, builds **sf** point maps and state summaries, and writes figure PNGs.

Primary input

- `august_combined_data.csv` (frozen manuscript dataset; path in script is machine-specific).

Note: This script does **not** read the fatality CSV; fatality analyses use the files in sections 1–4.

Ancillary data (retrieved when the script is run)

- `rnatrlearnth / rnaturalearthdata`: U.S. state boundaries.
- `tidycensus::get_acs`: state total population (B01003_001; ACS year is set in the script).

Outputs

- **roads_for_stats.csv** — community-level table after derivations (written to the working directory when the script is run).
- **Figures** (ggsave to `paper-figures/`), including: `map_hazard.png`, `map_vulnerability.png`, `map_exposure.png`, `map_risk.png`, `map_biv_risk.png`, `plot_num_communities.png`, `plot_population.png`, `plot_percent_state_pop.png`, `key_biv_risk.png`.
- Additional **ggplot** objects printed to the device (e.g. bivariate key).

Software

R	tidyverse, sf, rnaturalearth, rnaturalearthdata, RColorBrewer, scales,
packages	tidycensus, ggpubr, biscale
R version	<i>Insert tested version at submission.</i>
Other	Census API key for tidycensus (per package documentation); network access when fetching ACS and base map layers.

File dependencies (conceptual)

```
fire_fatalities_with_notes.xlsx
      |
      v (manual or workflow export)
  fire-fatality-for-R.csv
    /                   \
fatalities.R      fire_fatalities_supplement.R
```

```
august_combined_data.csv
      |
      v
    pnas-mapping.R
```